Letter Sequences & Alphabet Code (Set B)

Year 5 Accelerated · **Homework Worksheet**

Worked Example

Example — Mixed Code Practice: What comes next? C, F, I, L, __ and if A=1, B=2..., what number represents the letter in your answer?

Answer: O comes next (gap of +3 throughout: C, F, I, L, O). O is the 15th letter, so the number is 15.

Quick Practice (15 questions)

1. What comes next? B, D, F, H, __
2. What comes next? W, U, S, Q, __
3. If A=1, B=2, C=3..., what number stands for the letter J? A) 9 B) 10 C) 11 D) 12 E) 8
4. Select ALL that apply: which of these pairs of letters are exactly 3 letters apart? A) B and E B) F and I C) K and L D) N and Q E) R and S
5. What comes next? A, D, G, J, __
6. In a code where A=1, B=2, C=3..., which letter stands for the number 20? A) R B) S C) T D) U E) V
7. What comes next? Y, V, S, P, __
8. Fill in the missing letter: B, _, F, H, J (skipping one letter each time)
9. If every letter is shifted forward by 2 places (A→C, B→D...), what does E become? A) C B) D C) F D) G E) H
10. Select ALL that apply: in the sequence D, G, J, M, P, which statements are true? A) The gap between each letter is 3. B) The sequence decreases. C) The next letter after P is S. D) The first letter is D. E) The gap between J and M is 4.
11. What comes next? E, G, I, K, __
12. In the alphabet-position code (A=1...), what number represents the letter N? A) 13 B) 14 C) 15 D) 12 E) 16
13. What comes next? T, R, P, N, __
14. If letters are shifted BACK by 1 place (B→A, C→B...), what does G become? A) E B) F C) H D) I E) D
15. Select ALL that apply: which letters come BEFORE J in the alphabet? A) G B) M C) C D) L E) F

Mixed Practice (15 questions)

1. What comes next? A, C, F, J, __ (the gap increases by 1 each time: +2, +3, +4...)
2. A code shifts every letter forward by 3 places (A→D, B→E...). What does the word "DOG" become?
3. What comes next? B, E, I, N, __ (gaps increase by 1 each time: +3, +4, +5...)
4. In a code, A=1, B=2... Two numbers are added and converted to a letter: if the numbers are 9 and 10, which letter results? A) R B) S C) T D) U E) Q
5. Select ALL that apply: which sequences correctly keep a GAP of 5 between each letter? A) A, F, K, P, U B) B, G, L, Q, V C) C, H, M, R, X D) D, I, N, S, X E) A, E, I, M, Q
6. What comes next? Y, V, S, P, __ (gap is always -3) — give the next TWO letters.
7. A code shifts every letter BACK by 4 places (E→A, F→B...). What does the word "JQM" decode to?

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8. What comes next in this interleaved sequence? A, Y, C, W, E, __

9. In the position code (A=1…Z=26), what letter corresponds to 26 − 10? A) O B) P C) Q D) N E) R

10. Select ALL that apply: in a +6 shift code (A→G, B→H…), which of these are correctly matched? A) A→G B) D→J C) S→Y D) U→A (wrapping) E) Z→F (wrapping)

11. What comes next? D, G, K, P, __ (gap increases by 1 each time: +3, +4, +5…)

12. A code uses "position number + 4, then convert back to a letter" (wrap after Z). What does the letter X become?

13. What comes next? N, K, H, E, __

14. Select ALL that apply: in the sequence F, I, L, O, R, which of the following are correct? A) Each gap is +3. B) The sequence is decreasing. C) The next letter after R is U. D) F is the first term. E) The gap between L and O is 2.

15. In a shift code where every letter moves forward by 6 places, what does the letter N become?

Standard / Advanced (15 questions)

1. What comes next in this sequence, and explain the rule: C, D, F, I, M, __

2. A code shifts every letter forward by 6 places, wrapping round after Z. Decode the word "UV".

3. In a position code (A=1…Z=26), a word's code number is the SUM of its letters' positions. What is the code number of the word "DOG"?

4. Select ALL that apply: for the sequence Y, V, R, M, __, which statements about the pattern are correct? A) The gaps are -3, -4, -5… B) The next term is 9 (the letter I). C) The sequence increases. D) The gap between V and R is 4. E) The gap between R and M is 6.

5. A code works by tripling each letter's position number, then converting back to a letter (wrapping round after Z as needed, using the remainder when dividing by 26, with 0 meaning Z). What does the letter C become?

6. What comes next in this sequence, which interleaves two alphabet sequences moving in opposite directions? B, Y, D, W, F, __, __

7. In a shift code, the letter H becomes O. Using the same shift, what would the letter W become (wrap round after Z)?

8. A puzzle gives this worked example of a code: CAT → FDW. Work out the shift used, then decode the word "LWS" using the same shift.

9. Select ALL that apply: which of these correctly describe the sequence E, H, K, N, Q? A) Each position number increases by 3. B) The letters are every 3rd letter starting from E. C) As position numbers this is 5, 8, 11, 14, 17. D) The next letter after Q would be S. E) The gap is consistent throughout.

10. A code reverses the alphabet (A ↔ Z, B ↔ Y…). What does the word "DOG" become under this code?

11. What comes next in this sequence, and what is the underlying rule? B, C, E, H, L, __

12. In a position-number code where A=0, B=1, C=2… (one less than usual), what 3-letter word has the code 1, 13, 4?

13. A code shifts consonants forward by 1 place but leaves vowels (A, E, I, O, U) unchanged. Decode the effect on the word "DOG".

14. Select ALL that apply: a student claims the sequence V, S, P, M, J continues correctly with G as the next letter. Which statements support or refute this? A) The gap throughout is -3. B) V is the 22nd letter, so the next term should be 22 − (3×4) = 10, which is J, not G. C) J is the correct next letter. D) G is the correct next letter. E) The pattern must continue to G since it is "next" in a simple -3 pattern from J alone, ignoring the full sequence.

15. A two-step code first shifts every letter back by 1 place, then reverses the order of the letters in the word. Apply this to the word "DOG" and give the final 3-letter result.

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